

Requirements Table — Digital Campus Library Reservation Gateway

Exactly 5 Functional Requirements (FR-001–FR-005) and 2 Nonfunctional Requirements (NFR-001–NFR-002).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Student Member to search the catalog by ISBN, title, author, or keyword and return matching book records with real-time availability status (Available / On Loan / Reserved).	High	1. Given a valid 10 or 13-digit ISBN, the system returns the exact matching catalog record within 2 seconds. 2. Given a partial title/author keyword, the system returns all matching records ranked by relevance. 3. Given a non-existent ISBN, the system returns "No results found" with zero matches. 4. Each result shows availability status accurate as of the last completed transaction.	Students need a fast, reliable way to locate books and know availability before deciding to reserve or visit the library, reducing wasted trips and support-desk queries.
FR-002	Functional	The system shall allow a Student Member to place a reservation (hold) on a book that is currently unavailable, provided the member has not exceeded the maximum active reservation limit of 5 and does not already hold an active reservation for the same ISBN.	High	1. Given availability = "On Loan" and active reservations < 5, a reservation request succeeds and status becomes "Reserved". 2. Given a duplicate active reservation for the same ISBN, the request is rejected with a "Duplicate reservation" error. 3. Given 5 active reservations, the request is rejected with a "Maximum reservation limit reached" error. 4. A confirmation with a unique reservation ID and queue position is returned within 2 seconds.	Reservations let students secure high-demand titles fairly instead of relying on repeated manual checks, while limits prevent hoarding and ensure equitable access.
FR-003	Functional	The system shall maintain a First-In-First-Out (FIFO) waiting queue per ISBN, such that reservations are fulfilled strictly in the order placed, and each member can view their exact queue position.	High	1. Given N existing reservations on an ISBN, a new reservation is assigned position N+1. 2. When a copy becomes available, the reservation with the earliest timestamp is offered first. 3. If a member cancels, all subsequent members' positions decrement by 1 automatically. 4. Queue position matches timestamp-sort order with 100% consistency.	FIFO ordering guarantees fairness — the student who requested first is served first — a core expectation of any holds system and a way to prevent disputes.
FR-004	Functional	When a reserved copy becomes available for a member, the system shall automatically send an availability notification (email/in-app) within 5 minutes, and the hold shall remain valid for a 48-hour pickup window before expiring and passing to the next member.	High	1. Given a reservation reaches position 1 and a copy becomes available, a notification is dispatched within 5 minutes with title, deadline, and pickup location. 2. If not checked out within 48 hours, the hold expires and passes to the next member. 3. Notification delivery status is logged for every event. 4. No notification is sent to members further back in the queue while an earlier hold is active.	Timely notification with a bounded pickup window keeps the queue moving efficiently and prevents a copy being held indefinitely by an unresponsive member.
FR-005	Functional	The system shall support checking out and returning books, automatically calculate overdue fines for late returns at a fixed rate per day beyond the due date, and update the member's account and book availability accordingly.	High	1. On checkout, due date = loan date + 14 days; status changes to "On Loan". 2. On late return, a fine of ₹5/day is posted, capped at ₹200/item. 3. On return, status updates to "Available"/"Reserved" within 1 minute. 4. Members with fines > ₹500 are	Automating checkout/return and fine calculation ensures consistent policy enforcement, reduces manual librarian effort, and gives members transparent accountability for late returns.

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				blocked from new reservations/checkouts with a clear error message.	
NFR-001	Nonfunctional	The system shall respond to catalog search and reservation-status requests within 2 seconds under normal load (up to 500 concurrent users), and shall support at least 10,000 catalog records without response-time degradation.	High	1. Under 500 concurrent users, 95th-percentile search response time ≤ 2 seconds. 2. With 10,000+ catalog records, search response time ≤ 2 seconds for 95% of requests. 3. Throughput sustains ≥ 50 requests/second with error rate $< 1\%$.	Students and staff expect near-instant feedback during peak periods (e.g., start of semester); slow searches undermine adoption of the digital gateway over manual processes.
NFR-002	Nonfunctional	The system shall authenticate all users via campus SSO before allowing reservation, checkout, or account operations, encrypt all data in transit (TLS 1.2+) and at rest, and maintain 99.5% monthly uptime, logging all failed logins and reservation/checkout transactions.	High	1. Any reservation/checkout/account request without a valid session is rejected (HTTP 401). 2. All traffic verified to use TLS 1.2+. 3. Rolling 30-day uptime $\geq 99.5\%$ ($\leq \sim 3.6$ hrs downtime). 4. 100% of auth failures and reservation/checkout/return transactions appear in the audit log with timestamp, user ID, action.	As a system handling personal member data and library assets, it must protect privacy, prevent unauthorized account access, and remain dependably available during registration/exam periods.